

Laboratory 3: Induction Machine Braking & Generating

version 2.0

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Introduction

In this laboratory session you will investigate the operation of the induction machine above synchronous speed such that power flow is now **from** the mechanical system **to** the electrical load/grid. In the case of braking this is a temporary condition but even then, without additional braking circuitry included on the DC link this situation could lead to inverter device destruction due to excessive voltage on the DC Link (all semiconductor devices have a maximum voltage limit). Generating mode always operates the machine above synchronous speed caused by the external source of torque and power. In Exercise 3 you will investigate the advantages of variable speed generator control to maximum power/energy capture in a small wind turbine application.

Aims and Objectives

Model 1: Induction motor drive with assisting load

- Determining mechanical power in generating mode
- Effects on DC Link voltage when an assisting load is present

Model 2: DC Link Braking Resistor

- DC Link Braking Resistor topology
- DC Link Braking control
- Calculation of DC Link Braking Power

Model 3: Variable Speed Induction Generator

- Implementation of Aerodynamic model
- Implementation of mechanical model
- Fixed Speed operation and power curve
- Variable speed operation and power curve
- Limiting maximum output power



Model 1: Induction motor drive with assisting load

In this experiment we want to model a Sinusoidal PWM induction machine drive which is driven above synchronous speed by an assisting load. To save time an existing model (Lab3_Model1.bak) can be found on Moodle, see Appendix for instructions as to how to load this model into Portunus. The model should look like that shown on Figure 1.

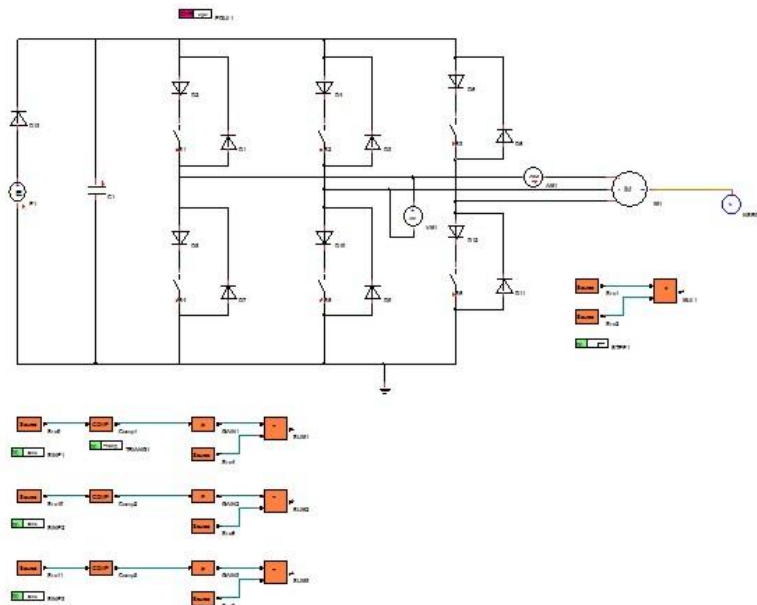


Figure 1: Sinusoidal PWM Induction Machine Drives

Set the various component parameters as outlined in Table 1:

Component Name	Component Type	Parameter Setup
E1	Voltage Source	Voltage – TR (V) = 580V
C1	Capacitor	Capacitance = 10,000e-6
IM1	Induction Machine	Stator Resistance = 5 Rotor Resistance = 3 Stator Leakage Inductance = 9.5e-3 Rotor Leakage Inductance = 9.5e-3 Main Inductance = 0.9 Number of Pole Pairs = 3 Moment of Inertia = 0.1
NSRC1	Speed Source	Speed (rpm) = 1020
SINE1	Sinewave	Frequency = 50 Amplitude = 1 Phase Shift = 0 Offset = 0
SINE2	Sinewave	Frequency = 50 Amplitude = 1 Phase Shift = 240 Offset = 0
SINE3	Sinewave	Frequency = 50 Amplitude = 1 Phase Shift = 120 Offset = 0
TRIANG1	Triangle Waveform	Frequency = 10e3 (note 10KHz PWM frequency now) Amplitude = 1 Phase Shift = 0 Offset = 0

Table 1: Model Setup

Exercise 1: Determining the DC Link voltage rise time under the assisting load condition.

Output the DC Link Capacitor Voltage (C1.V) onto an On Sheet Display. Also measure the induction machine shaft torque (MUL1.OUT)

Set $TEND = 1\text{ s}$ in the simulation parameter setup panel [F9] and run the simulation.

Parameter	Value
Shaft Torque (Nm) @ 700V DC Link	12.2
Estimated Mechanical Power (W)	1.303k
Time taken (t_{ch}) for DC Link Voltage to rise to 700V (t_{ch}) (s)	713.5ms

Note: Induction Machine Shaft Torque will increase over the time period due to the DC link voltage increase (and hence motor line voltages increase) – determine the time it takes for the DC link to reach **700V** and estimated from the graph the Shaft Torque at this time.

Lab Book Exercise: Verify this voltage rise time with appropriate equivalent circuit and DC Link energy **hand calculations**. Indicate any **approximations** included in the calculation.

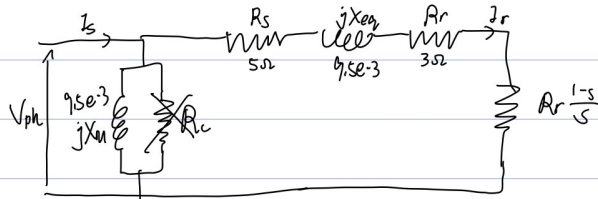
Modify the simulation model to determine the length of time it takes the DC Link voltage to rise to 700V (initial 580V) when the slip is now 0.04 (generating) – see Appendix for relevant slip equation when generating:

Parameter	Value
Machine Speed (rpm)	1040
Shaft Torque (Nm) @ 700V DC Link	25.28
Mechanical Power (W)	2.753k
Time taken (t_{ch}) for DC Link Voltage to reach 700V (s)	352.89ms

Note: Shaft Torque will increase over the time period due to the DC link voltage increase (and hence motor line voltages increase) – determine the time it now takes for the DC link to reach **700V** and estimated from the graph the Shaft Torque at this time.

Comment on the change in charging time and the reasons behind this change:

The charging time is much shorter because in the generating case the shaft torque and speed at 700 V are higher, so the mechanical power delivered to the DC link roughly doubles ($\approx 1.3\text{ kW} \rightarrow \approx 2.75\text{ kW}$). Since the capacitor must gain the same amount of energy to rise from 580 V to 700 V, the higher input power charges it faster, so the DC link reaches 700 V in about half the time..



$$V_L = \frac{\sqrt{3}}{2\sqrt{2}} V_{dc} = \frac{\sqrt{3}}{2\sqrt{2}} \times 700 = 428.66 \text{ V} \quad (\text{Assump } V_{dc} = 700 \text{ in the worst case})$$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = 247.49 \text{ V}$$

$$N_s = \frac{n_{rat}}{p} = \frac{120 \times 50}{3 \times 2} = 1000 \text{ rpm}$$

$$s = \frac{N_r - N_s}{N_s} = 0.02$$

$$Z_T = R_s + \frac{R_r}{s} + jX_{eq} = 155.00 \angle 3.51^\circ$$

$$I_r = \frac{V_{ph}}{Z_T} = 1.5967 \angle -3.51^\circ$$

$$P_{mech} = 3 \cdot I_r^2 R_r \cdot \left(\frac{1-s}{s}\right) = 1.124 \text{ kW}$$

$$P_{loss} = 3 \cdot I_r^2 (R_s + R_r) = 61.19 \text{ kW}$$

$$\Rightarrow P_{DC} = P_{mech} - P_{loss} = 1.063 \text{ kW}$$

$$E_{580} = \frac{1}{2} C V^2 = \frac{1}{2} \times 10^4 \mu \times 580^2 = 1682 \text{ J}$$

$$E_{700} = \frac{1}{2} C V^2 = \frac{1}{2} \times 10^4 \mu \times 700^2 = 2450 \text{ J}$$

$$\text{therefore, } t_{ch} = \frac{E_{700} - E_{580}}{P_{DC}} = 722.5 \text{ ms}$$

Model 2: DC Link Braking Resistor

This over-voltage condition could clearly be a problem if the DC Link voltage exceeded the maximum voltage of the semiconductor switches. Also, in applications where braking is required (eg traction) the DC Link is again charged up during the braking period. One possible solution to ensuring that the DC Link voltage is not allowed to exceed a given value is to include a DC Link Braking Resistor (+ associated switch) as shown on Figure 2.

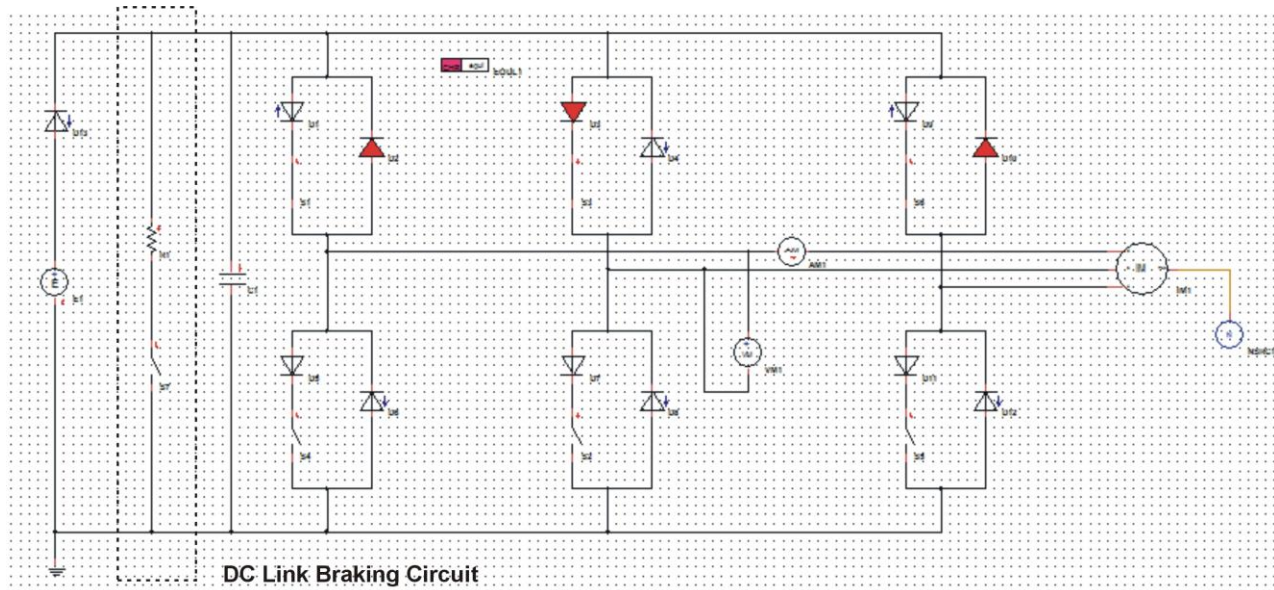


Figure 2: DC Link Braking Circuit

The simplest form of control for the DC Link Braking switch is a Hysteresis controller whereby the switch is turned ON when the DC Link voltage is above an upper Threshold (V_{Th2}) causing the DC Link power to be dissipated in the resistor. The switch then remains on until the DC Link voltage falls below a lower threshold (V_{Th1}) at which point the switch is turned OFF and the DC Link voltage once again rises due to the braking power from the induction machine. This operation is outlined on Figure 3. The rate at which the DC Link voltage falls during switch ON is dependant on a) the mechanical power of the induction machine, and b) instantaneous power dissipated in the load resistor. Also the MAXIMUM continuous power the resistor can dissipate is when the switch is always ON which equals V_{dc}^2/R_{Brk} – so I would suggest you choose the value of braking resistor such that this maximum power is **twice** the maximum mechanical power from the induction machine (see calculation below) to ensure that the braking power will reduce the DC link voltage at a rate of change similar to charging period.

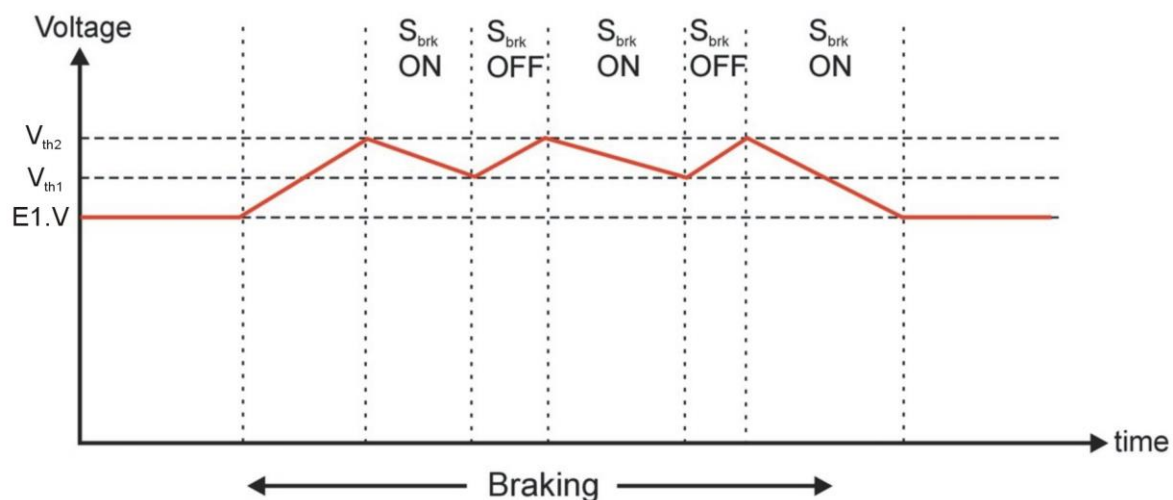


Figure 3: Hysteresis Braking Control

Calculation of Braking Resistor value given $V_{Th1} = 630V$ and $V_{Th2} = 650V$ and:

$$P_{Brk} = \frac{V_{dc}^2}{R_{Brk}}$$

Step	Calculation	Value
1	Mechanical Power @1040rpm (see previous results)	2.753kW
2	$P_{Brk} = \text{TWICE Mechanical Power}$	5.506kW
3	Average DC Link Voltage during braking (V_{dc})	640V
4	Braking Resistance value R_{Brk}	74.39 Ohm

You now need to consider how to actually implement the Braking Switch Hysteresis Controller in Portunus. Using the model you modified in Exercise 1 as the starting point add the DC Link resistor/switch and also include the **Two Point /Hysteresis** model in the **Blocks** directory and set up parameters as outlined in Figure 4/Table 2.

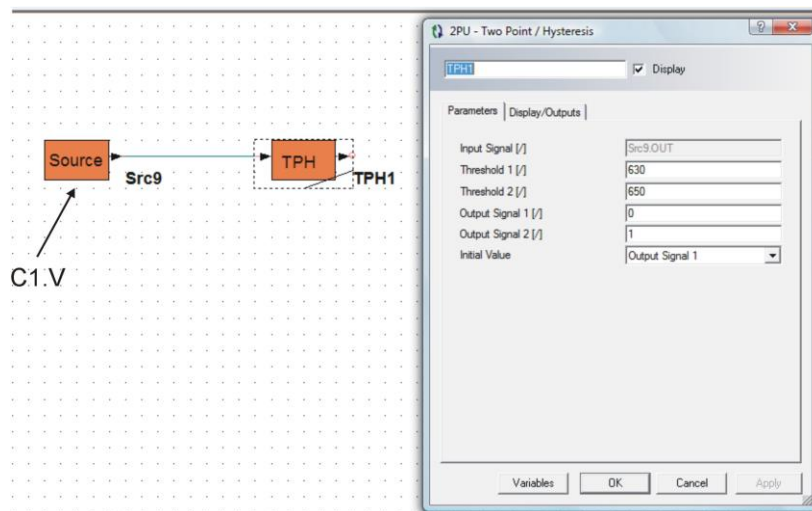


Figure 4: Portunus Hysteresis Controller

Parameter	Value
Threshold 1 (V)	630
Threshold 2 (V)	650
Output Signal 1 (V)	0
Output Signal 2 (V)	1
Initial Value	Output Signal 1

Table 2: Two Point Hysteresis Component Setup

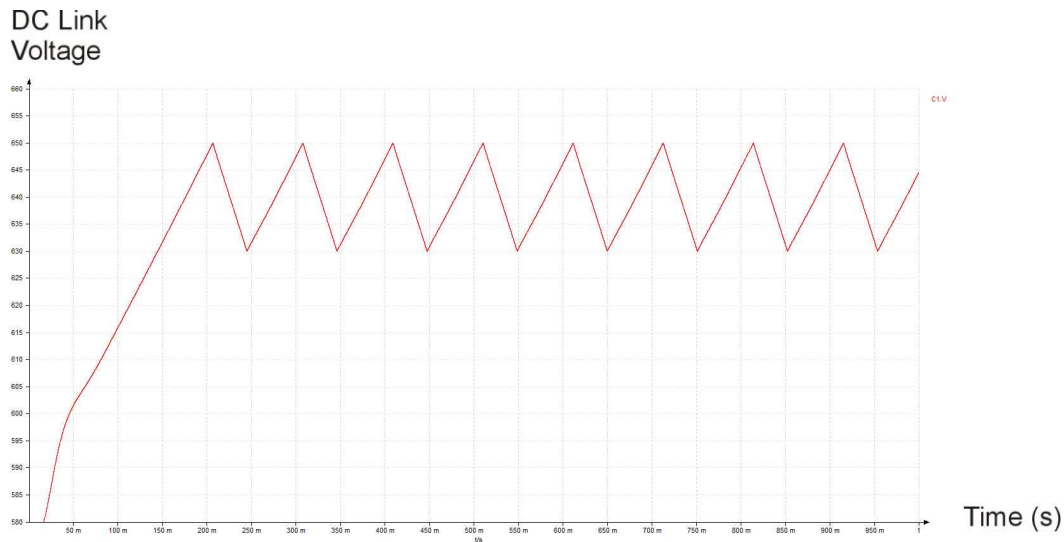
Allocate TPH1.OUT to the braking switch (S7) **Control Signal** Parameter.

Set **NSRC1** Speed Parameter = 1040rpm

Output the DC Link Voltage (C1.V) onto an On Sheet Display (should be there already!)

Set **TEND** = 1s in the simulation parameter setup panel [F9] and run the simulation.

Sketch, print or photograph the resultant DC Link Capacitor Voltage (C1.V) and determine the switching period/frequency (measure this as accurately as you can as you will use this period in the next section to determine average power dissipated in the resistor!!!):



Estimated Braking Switching Period (s)	$T = 712.134 - 610.858 = 101.276\text{ms} = 0.101\text{ s}$
Switching Frequency (Hz)	9.874

Exercise 2: Measurement of Average Power Dissipated in Braking Resistor

Implement the necessary blocks within Portunus to determine the average power dissipated in the Braking resistor. We do this (as before) by multiplying braking resistor voltage and current to get instantaneous braking power, then using a Signal Analyser (Sana) to measure AVERAGE Braking Power. Note that the Sliding Window of the Signal Analyser should be set to the **Estimated Braking Switching Period** determined above.

We also want to compare this average braking resistor power with the average mechanical power generated during this period. To ensure as accurate an estimation of this as possible the machine torque should be averaged as well using a Signal Analyser block with a sliding window also set to the **Estimated Braking Switching Period**.

Set $TEND = 1\text{ s}$ in the simulation parameter setup panel [F9] and run the simulation.

Determine the following through simulation/calculation:

Machine Speed (rpm)	Switching Period (τ) s	Average Machine Torque (Nm)	Mech Power (W)	Braking Resistor Power (W)	Δ Power (W)
1040	0.101	19.488	2122.41	2090.91	31.5

Comment on the results and suggest possible reasons for the difference between Mech Power and Braking Resistor Power.

The braking resistor power is slightly lower than the mechanical power because not all of the generated mechanical energy is dumped into the resistor. Part of it is lost as copper and core losses in the machine, friction and windage, and switching/conduction losses in the converter. There is also some error from using an estimated switching period and averaging over a finite window, so the measured powers will not match exactly.

Model 3: Variable Speed Induction Generator

In this final model we will explore the advantages, in terms of energy production, of a variable speed induction generator compared with the traditional fixed speed approach. The application is a **13kW** wind turbine the model for this (Lab3_Model3.bak) can be downloaded from Moodle as outlined earlier. The complete wind turbine model includes the following components:

- Aerodynamic model producing mechanical power and torque for a given wind speed and turbine speed. The turbine has 6.5m radius blades
- Mechanical model including an **18:1** gearbox between the blades (low speed) and induction machine (high speed)
- **6 pole 11kW** Induction machine model
- 3 phase Inverter
- DC Link Braking circuit

Have a good look at how each of these components is implemented in Portunus and satisfy yourself that the aerodynamic equations correspond to those presented in the lecture. The Aerodynamic model calculates turbine torque at the given turbine speed and wind speed and this torque is then used to drive the induction machine above synchronous speed.

Exercise 3.1 Fixed Speed Operation

To analyse this mode of operation you need to set the generator synchronous speed at 1000rpm (Sine Control = 50Hz, Ma=1) and then set the Wind Speed input (Src17) to the values shown on the following table and from the simulation (outputting the necessary parameters to On Sheet Displays) determine the induction Machine shaft power (MUL5.OUT) and the corresponding induction machine speed (IM1.N) and Cp (CHR1.OUT) at each wind speed. Also output the DC Link capacitor voltage (C1.V) to ensure that the braking controller is operating correctly throughout the series of tests.

Set the **Initial Speed** parameter in the Induction Machine = 1000rpm

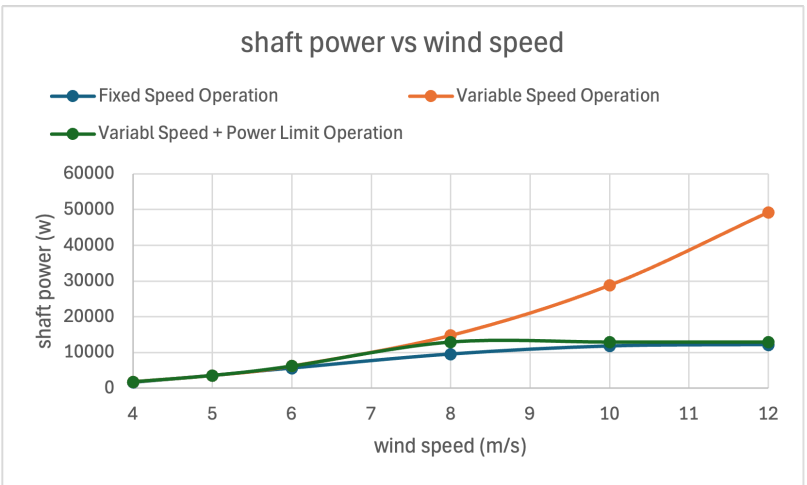
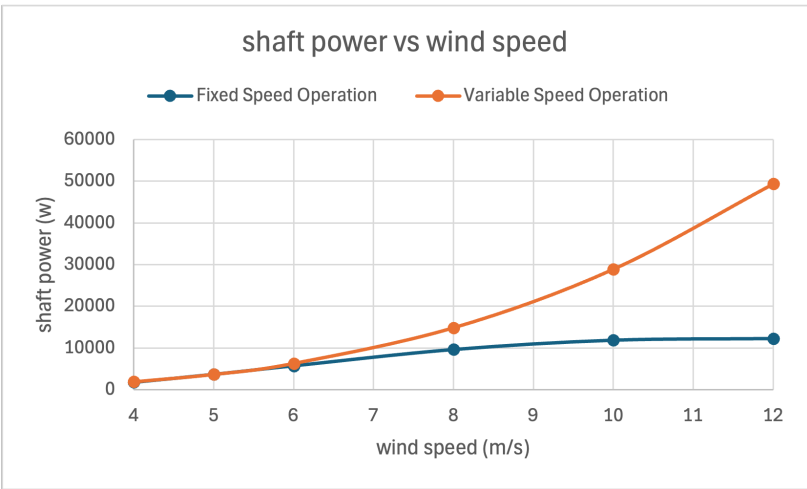
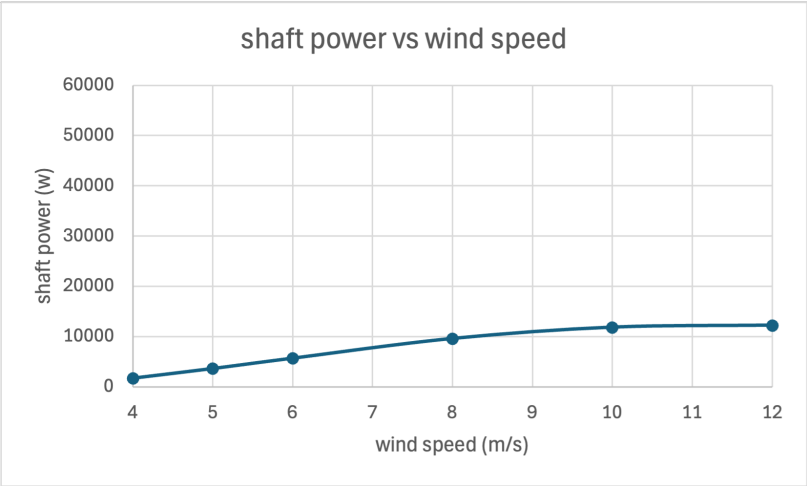
Set **TEND = 0.5S** in the simulation parameter setup panel [F9] and run the simulation.

Wind Speed (m/s)	Induction Machine Speed (rpm)	Cp	Induction Machine Shaft Power (W)
4	1004.85	0.334	1702.7
5	1010.04	0.363	3617.1
6	1015.53	0.329	5667.12
8	1025.63	0.235	9578.39
10	1031.36	0.148	11819.3
12	1032.33	0.089	12213.4

Plot a graph of shaft power (NOTE: make maximum power on y axis = 60kW) against wind speed (x axis) to determine the power curve for this machine operating at fixed speed (note the speed does change slightly due to the different torque at each wind speed). Have a look at the Cp Look Up Table (XY2) where the x input is Tip Speed Ratio (λ) and the output f(x) is Cp. Using this and the results above estimate the wind speed at which Maximum Cp occurs.

Wind Speed at which Maximum Cp occurs	4.97m/s
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tip speed ratio = 7.689
 $v = 1010.04$
 $r = 6.5$
 wind speed = 4.97
 $C_p = 0.3643$



Exercise 3.2 Variable Speed Operation

The induction machine is now to operate as a variable speed drive by controlling the synchronous frequency of the inverter therefore allowing the turbine to operate at Maximum C_p at any wind speed.

Calculate the corresponding induction machine speed to operate at Maximum C_p at the following wind speeds and by controlling the synchronous frequency of the inverter simulate at these operating points to determine induction machine shaft Power and corresponding induction machine speed and C_p .

For every test point set the **Initial Speed** parameter in the Induction Machine = synchronous speed (eg 1000rpm for 50Hz)

Wind Speed (m/s)	Synchronous Frequency (Hz)	Induction Machine Speed (rpm)	C_p	Induction Machine Shaft Power (W)
4	40.666	813.318	0.3641	1856.02
5	50.832	1016.647	0.3640	3623.62
6	60.999	1219.977	0.3638	6258.11
8	81.332	1626.636	0.3631	14807.3
10	101.665	2033.295	0.3621	28835.7
12	121.998	2439.954	0.3581	49285.4

(Note – for simplicity keep $M_a=1$ for all values of synchronous frequency)

Draw the resultant shaft power on the same graph as fixed speed option.

Exercise 3.3 Variable Speed + Power Limit Operation

The final control requirement is that the maximum output power must NOT be allowed to exceed **13kW** at any wind speed in this range due to the induction machine being rated at 13kW maximum continuous operation – so using the simulation model (and/or the C_p curve shown on Figure 5 and relevant equations) determine the corresponding change in synchronous frequency at the speeds at which $>13kW$ output occurs to limit the output to 13kW.

Wind Speed (m/s)	Synchronous Frequency (Hz)	Induction Machine Speed (rpm)	C_p	Induction Machine Shaft Power (W)
4	40.666	813.318	0.3641	1856.02
5	50.832	1016.647	0.3640	3623.62
6	60.999	1219.977	0.3638	6258.11
8	62.408	1248.167	0.3123	12996.8
10	52.889	1057.77	0.1600	12993.3
12	53.946	1078.92	0.0925	12992.8

Draw the resultant shaft power on the same graph as fixed speed option.

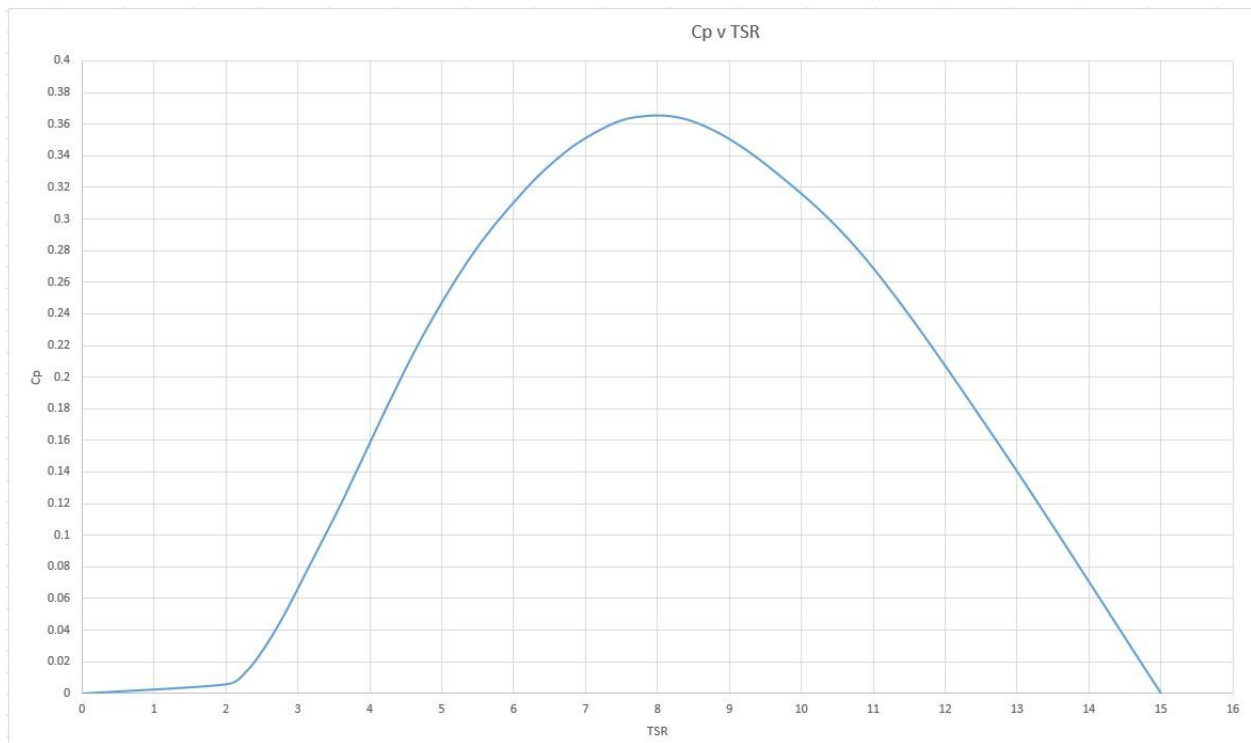


Figure 5: Cp vs TSR

$$Tip_Speed_Ratio(TSR) = \frac{Tip_Speed}{v} = \frac{\omega.R}{v}$$

$$Mech_Power(W) = \frac{1}{2} \cdot \rho \cdot \pi \cdot R^2 \cdot C_p \cdot v^3$$

where:

ω = Turbine speed (rad/s)

R = Turbine blade radius (6.5m)

v = wind speed (m/s)

ρ = Air Density (typ 1.225kg/m³)

C_p = Coefficient of Power

Appendix

Accessing Existing Models

In this laboratory session you are required to access an existing Portunus models (Lab3_Model1.bak and Lab3_Model3.bak) from Moodle page ENG4187. Copy these files into a known directory on your PC. To open this file in Portunus select 'File-Open' and go to the directory where you have stored the files. Note that in Portunus you should select 'all files' in the **Open File** panel to access the .bak file. If you subsequently want to save models then choose the relevant Lab3_Model'X'.**bak** file as it is MUCH smaller than the corresponding .ecd file but contain all the necessary information and can be subsequently opened from Portunus using the select 'all files' in the **Open File** panel.

Slip Equations:

Motoring Condition (actual speed N_r below synchronous speed N_s):

$$Slip(s) = \frac{N_s - N_r}{N_s}$$

Braking or Generating Condition (actual speed N_r above synchronous speed N_s):

$$Slip(s) = \frac{N_r - N_s}{N_s}$$